**Name:**

**Student ID:**

**Module 2 Homework - CEC 300: Computing for Aviation and Aerospace**

**Expected Completion Time: 2-4 hours**

**Problem 1:** Write a small application in Matlab, Python or Java to simulate the analysis of DME pulses.

The pulses are measured relative to an unsynchronized clock. The program will take in four(4) numbers representing the time (in µs) of the leading edge of each pulse . The first two pulses are the transmitted pulses from the aircraft to the DME station, and the other two pulses correspond to the response as received by the airplane.

e.g. DME(1862, 1874, 2024, and 2036)

Has a pulse pair spacing of 1874-1862 = 12 µs and 2036-2024 = 12 µs. This would be a Channel X signal.

The time between first pulses of the sent and received signals is 2024-1862 = 162 µs.

We determine the distance as (162 – 50 µs)/(12.36 s/nmi)= 9.06 nmi (nautical miles), where 50 µs is the delay for Channel X.

**<u>Task:</u>** Create a Matlab, Python, Java, or C function that takes the time of 4 pulses. Use only channel X and Y.

- If correct pulses for channel X are detected the system will print
  - **Channel is X, distance is dd.d nautical miles.**
    - Where dd.d is the correct distance such as 12.2.
- If correct pulses for channel Y are detected the system will print
  - **Channel is Y, distance is dd.d nautical miles.**
    - Where dd.d is the correct distance such as 12.2.
- If the system receives a valid pulse spacing for the X channel when transmitting on the Y channel or vice versa, this means that we have received pulses from an aircraft other than our own. In this case, the system will print:
  - **FRUIT detected.**
- If the measured time between the transmitted and received pulses exceeds 200 nmi than the system will print (depending on channel):
  - **Channel is X, no reply.**
  - **Channel is Y, no reply.**
- If incorrect pulses spacings (receive or transmit) for channel X or Y are detected **<u>or</u>** if the received signals arrive in less than the built-in delay of the ground station print
  - **Signal Garbled.**

**Note:** To allow proper grading, the function shall be implemented so the arguments look like the example below, and the result shall be displayed in the following command line. Submit your source code with your algorithm, your main method, and a readme file providing me with compilation/run instructions. Please submit your files in a .zip file with the format "cec300-m2-yourlastname.zip."

**Grading (for Problem 6)**

| | |
|---|---|
| System identifies channel X and channel Y pulses correctly | 10 points |
| System calculates the correct distance. | 10 points |
| System correctly identifies FRUIT. | 10 points |
| System correctly identifies garbled signals | 10 points |
| System correctly handles no reply condition | 10 points |
| Total | 50 points |

Example:
>> DME(572,584,772,784)
Channel is X, distance is 12.14 nm.

You can test your code with a wide range of test cases. Some sample test cases are below. There is no order in these test cases, nor they cover all the scenarios previously described.

DME(572,584,772,784)
DME(1011,1047,1361,1391)
DME(1671,1683,2081,2117)
DME(2307,2319,2337,2349)
DME(5181,5217,7767,7797)
DME(8011,8047, 8300,8322)

**Problem 2 [short answer]: Given the following VOR indicator, please answer the following questions.**

a. What radial is the VOR set to?

b. Does the indicator show that we must turn left or right?

c. What degree of heading change is needed (approximately) based on the indicator? How did you determine this?

**Problem 3 [short answer]: Answer the following questions about ILS?**

a. If your aircraft is left of the runway center line before landing, what would be the characteristics of the signal received by the aircraft? How would the signal change if the aircraft is right of the runway?

b. Which ILS category would you a transport category jet flying arriving at an airport with 950 feet of runway visibility?

**Problem 4 [essay]: Answer the following questions about GPS. Your response should be at least 500 words.**

In your own words, explain the process of how a GPS receiver determines its three-dimensional position and discuss the various sources of error that can affect the accuracy of this calculation.

**Problem 5: Answer the following questions about satellite sensors and actuators.**

a. Compare and contrast reaction wheels vs. magnetorquers. How do they complement each other?

b. Explain in your own wods how fiber-optic gyroscopes are used to determine the angular rate about an axes. Your answer should include a brief discussion of the Sagnac Effect.

c. How does the Deep Space Network enable satellite guidance and control for spacecraft operating beyond cislunar space?